

Def. Let G be a group, and $H \leq G$ be a subgroup.
 Given an element $g \in G$, define left-Coset of H as

$$gH \stackrel{\text{def}}{=} \{gh \mid h \in H\}.$$

Lemma. Let G be a group, and $H \leq G$ be a subgroup. Then, for each $g_1, g_2 \in G$,
 $g_1H = g_2H$ if and only if $g_1^{-1}g_2 \in H$.

Proof. Given $g_1, g_2 \in G$, suppose that $g_1H = g_2H$. Then, since $g_2 \in g_2H = g_1H$ ($1 \in H$),
 $g_2 = g_1h$ for some $h \in H$, thus $g_1^{-1}g_2 \in H$.

Conversely, let $g_1, g_2 \in G$ be given st $g_1^{-1}g_2 \in H$. That is, $g_1h = g_2$ for some $h \in H$.
 Given $g_1h_1 \in g_1H$, $g_1h_1 = g_2h^{-1}h_1 \in g_2H$. Given $g_2h_2 \in g_2H$, $g_2h_2 = g_1hh_2 \in g_1H$. $\square$

Q. In left Coset sense, $g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H$ hold?

A. No! Ss, $H = \langle (1, 2) \rangle$, $g_1 = (1\ 3)$ and $g_2 = (1\ 2\ 3)$

Thm. Lagrange Thm

If H is a subgroup of a group G , then $|G|/|H| = [G:H]$

Proof. Given a subgroup $H \leq G$, define a relation on G as;

$$x \sim y \Leftrightarrow xH = yH$$

Then, it is an equivalence relation, so a Collection $\{gH \mid g \in G\}$ forms a partition.
 Moreover, for each $g \in G$, a map

$$h \mapsto gh \quad (h \in H)$$

is onto by definition and one to one by the cancellation law, so $|H| = |gH|$.

Isomorphism Theorem.

1st Isomorphism theorem.

For any Homomorphism $\varphi: G \rightarrow G'$,

$$G/\ker \varphi \cong \text{Im } \varphi.$$

Proof. Since $\ker \varphi \trianglelefteq G$, $G/\ker \varphi$ is well-defined as a group. Define

$$\tilde{\varphi}: G/\ker \varphi \rightarrow \text{Im } \varphi: g/\ker \varphi \mapsto \varphi(g).$$

Then, $g_1/\ker \varphi = g_2/\ker \varphi \Leftrightarrow g_1 g_2^{-1} \in \ker \varphi \Leftrightarrow \varphi(g_1) = \varphi(g_2)$

so $\tilde{\varphi}$ is well-defined and injective, and the surjective is clear. $\square$

2nd Isomorphism theorem.

Let G be a group, and let A, B be subgroups of G satisfying $A \trianglelefteq N_G(B)$.

Then, $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$. Further,

$$AB/B \cong A/A \cap B$$

Proof. Since A normalizes B ,

$$AB = \bigcup_{a \in A} aB = \bigcup_{a \in A} Ba = BA$$

and so given $a_1 b_1, a_2 b_2 \in AB$, $a_1 b_1 (a_2 b_2)^{-1} = a_1 b_1 b_2^{-1} a_2^{-1} \in AB$, so $AB \leq G$.

$B \leq AB$ is trivial, and given $ab \in AB$ and $g \in B$, $abgb^{-1} \in B$, so $B \trianglelefteq AB$.

$A \cap B \trianglelefteq A$ is also trivial (These are given by the fact that A normalizes B).

Now, define a map

$$\varphi: A \rightarrow AB/B: a \mapsto aB.$$

It is well-defined since $A \leq AB$ and it is Homomorphism because: given $a_1, a_2 \in A$,

$$\varphi(a_1 a_2) = a_1 a_2 B = a_1 B \cdot a_2 B$$

and onto: given $abB \in AB/B$, take $a \in A$, then $\varphi(a) = aB = abB$.

And, the kernel is $\ker \varphi = \{g \in A: gB = B\} = \{g \in A: g \in B\} = A \cap B$. Consequently, the first isomorphism theorem gives $A/A \cap B \cong AB/B$, $\square$

3rd Isomorphism Theorem.

Let G be a group and $H, K \trianglelefteq G$ be normal subgroups with $H \leq K$.

Then, $K/H \trianglelefteq G/H$ and $[G/H]/[K/H] \cong G/K$.

Proof. Since $K \trianglelefteq G$, $K/H \trianglelefteq G/H$ directly. Define a map

$$\varphi: G/H \rightarrow G/K: gH \mapsto gK.$$

It is well-defined, because: given $g_1, g_2 \in G$,

$$g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H \Rightarrow g_1g_2^{-1} \in K \Leftrightarrow g_1K = g_2K = \varphi(g_1) = \varphi(g_2)$$

$\uparrow$
 $H \leq K$ (so, injectiveness is given only $H=K$).

Surjectivity and Homomorphism are clear. Moreover,

$$\ker \varphi = \{gH \in G/H \mid gK = K\} = \{gH \in G/H \mid g \in K\} = K/H.$$

Finally, the 1st isomorphism theorem gives the result.

